

Simple Interest vs Compound Interest

Simple Interest can be calculated by $(\text{principal}) \times (\text{rate}) \times (\text{time})$

Formula: $I = prt$

Example:

Find the interest earned by investing \$2,400 at 3% interest rate for 5 years.

$$I = prt$$

$$I = (2400)(0.03)(5)$$

$$I = \$360$$

Find the future value of the investment after 5 years.

$$\text{Future Value} = \text{principal} + \text{interest}$$

$$\text{Future Value} = 2400 + 360$$

$$\text{Future Value} = \$2730$$

Compound Interest is calculated by the same formula, except that the interest is calculated at the end of each interest period and then the interest is added to the principal. Compound interest is calculated at the end of each subsequent investment period and the interest continues to be added to the principal.

Exercise 1

You put \$3000 in the bank at 2% APR that is compounded annually. How much money do you have in the account after 6 years? Finish the table to complete this problem.

Year	Beginning Balance	Interest Calculation	Interest	Ending Balance
1	3000	$(3000)(.02)$	60.00	3060.00
2	3060.00	$(3060.00)(.02)$	61.20	3121.20
3	3121.20	$(3121.20)(.02)$	62.42	3183.62
4	3183.62			
5				
6				

What would be the future value of the investment if *simple interest* was paid?

Exercise 2

You put \$4000 in the bank at 3% APR that is compounded annually. How much money do you have in the account after 6 years?

NOTE: We'll shorten the process by finding 103% of the principal at the end of each investment period.

Month	Beginning Balance	Ending Balance Calculation	Ending Balance
1	4000	$(4000)(1.03)$	4120.00
2	4120.00	$(4120)(1.03)$	4243.60
3	4243.60	$(4243.60)(1.03)$	4370.91
4	4370.91		
5			
6			

Exercise 3

1. You inherit \$120,000 from your rich Uncle Bucks. If you invest the money in a CD that pays 4% interest, how much money will you have in four years?

a. Compute the problem using simple interest. Show work. $I = prt$

b. Compute the problem if the interest is compounded annually.

Year 1: $(120,000)(1.04) = 124,800$

Year 2: $(124,800)(1.04) =$

Year 3:

Year 4:

2. Say you put \$10 in a savings account and the account pays 5% *simple interest* for each year that the money is in your account.

- a. Write a formula that could be used to determine the value of your account after x number of years.

$$f(x) =$$

- b. Describe the graph of the function.

- c. Use the formula to find the value of the account after 24 years.

$$f(24) =$$

3. Say you put \$10 in a savings account and the account pays 5% interest for each year the money is in the account but the interest is *compounded annually*.

- a. What would be the value of the account after 4 years?

- b. Describe the graph of the function.

- c. Write a formula that could be used to determine the value of the account after x number of years.

$$f(x) =$$

- d. Try the formula to find the value of the account in 4 years. Then, try the formula to find the value for 24 years.

Future Value for Compound Interest

$$FV = P(1 + R)^T$$

Exercise 4

Show work for each problem using the future value formula. Give your final answers to the *nearest cent*.

1. You win a \$1000 in a radio contest. If you put the \$1000 into an account that pays 3.5% APR *compounded annually*, what will be the value of your investment after 20 years?
2. Remember the \$120,000 that you inherited from your rich Uncle Bucks? If you invest the money in a CD that pays 4% interest *compounded annually*, how much money will you have in forty years?

Exercise 5

Label each graph to identify whether the graph could represent a *simple interest* problem or a *compound interest* problem.

